DAV Lab 1

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In [ ]:
```
my_int=10
my_float=3.14
my_string="Hello, World!"
my_bool=True
my_complex=2+3j

print(f"Integer:,{my_int}, { type( my_int ) }")
print(f"Float:,{my_float}, { type( my_float ) }")
print(f"String:, {my_string},{ type( my_string ) }")
print(f"Boolean:, {my_bool},{type( my_bool )}")
print(f"Complex:, {my_complex}, {type(my_complex)}")
```

In [2]:
```
my_int=20
print(my_int)
```

```
20
```

Strings

In [3]:
```
str1='Single quotes'
str2="Double quotes"
str3="""Triple quotes"""
print(str1)
print(str2)
print(str2)
str4=str1+" "+str2+" "+str3
print(str4)
```

```
Single quotes
Double quotes
Double quotes
Single quotes Double quotes Triple quotes
```

String Slicing

In [4]:
```
word="Python"
print(f"1st:",word[0])
```

```
print(f"Last:",word[-1])
print(f"Slice:",word[1:4])
print(f"Length:",len(word))
```

```
1st: P
Last: n
Slice: yth
Length: 6
```

Boolean

In [5]:
```
is_active=True
is_admin=False

print(f"Is Active: {is_active}, Type: {type(is_active)}")
print(f"Is Admin: {is_admin}, Type: {type(is_admin)}")
```

```
Is Active: True, Type: <class 'bool'>
Is Admin: False, Type: <class 'bool'>
```

In [6]:
```
result=( 10>5 )
print(f"is 10 greater than 5? {result}")
result=( 10==5 )
print(f"is 10 equal to 5? {result}")
```

```
is 10 greater than 5? True
is 10 equal to 5? False
```

Type Casting

In [7]:
```
n1_int=10
n1_float=float(n1_int)
print(f"Integer to Float: {n1_float}, Type: {type(n1_float)}")

n2_float=12.7
n2_int=int(n2_float)

print(f"Float to Integer: {n2_int}, Type: {type(n2_int)}")

num_str=str(123)
print(f"Integer to String: {num_str}, Type: {type(num_str)}")

str_num="45"
int_num=int(str_num)
print(f"String to Integer: {int_num}, Type: {type(int_num)}")
```

```
Integer to Float: 10.0, Type: <class 'float'>
Float to Integer: 12, Type: <class 'int'>
Integer to String: 123, Type: <class 'str'>
String to Integer: 45, Type: <class 'int'>
```

Boolean Type Casting

In [8]:
```
bool_true=bool(1)
bool_false=bool(0)
bool_str_true=bool("abc")
bool_str_false=bool("")

print(f"Bool(1): {bool_true}")
print(f"Bool(0): {bool_false}")
print(f"Bool('abc'): {bool_str_true}")
print(f"Bool(''): {bool_str_false}")
```

```
Bool(1): True
Bool(0): False
Bool('abc'): True
Bool(''): False
```

Conditional Statements

In [9]:
```
x=10
if x>5:
    print(f"{x} is greater than 5")

y=3
if y%2==0:
    print(f"{y} is even")
else:
    print(f"{y} is odd")

score=85
if score>=90:
    print(f"Grade:{score}\nA")
elif score>=80:
    print(f"Grade:{score}\nB")
elif score>=70:
    print(f"Grade: {score}\nC")
else:
    print(f"Grade: {score}\nD")
```

```
10 is greater than 5
3 is odd
Grade:85
B
```

In [10]:
```
age=15
has_license=True
if age>=18 :
    if has_license:
      print("You can drive.")
    else:
        print("You cannot drive.")
else:
```

```
    print(f"Age {age} is less than 18")
```

```
Age 15 is less than 18
```

Loops

In [11]:
```
fruits=['apple','banana','mango']
for i in fruits:
    print(i)
```

```
apple
banana
mango
```

In [12]:
```
word="Python"
for i in word:
    print(i)
```

```
P
y
t
h
o
n
```

In [13]:
```
for i in range(2,10,2):
    print(i)
```

```
2
4
6
8
```

In [14]:
```
count=0
while count<5:
    print(f"Count: {count}")
    count+=1
```

```
Count: 0
Count: 1
Count: 2
Count: 3
Count: 4
```

In [3]:
```
secret=7
guess=0
while True:
    guess+=1
    if guess==secret:
        print("Congratulations! You guessed the secret number.")
        break
    elif guess>secret+2:
        print("Too many wrong guesses!")
        break
    else:
        print(f"Guess: {guess}: Not correct...")
```

```
Guess: 1: Not correct...
Guess: 2: Not correct...
Guess: 3: Not correct...
Guess: 4: Not correct...
Guess: 5: Not correct...
Guess: 6: Not correct...
Congratulations! You guessed the secret number.
```

In [4]:
```python
num=0
while num<10:
    num+=1
    if num%2==0:
        continue
    print(f"Odd Number: {num}")
```

```
Odd Number: 1
Odd Number: 3
Odd Number: 5
Odd Number: 7
Odd Number: 9
```

Functions

In [5]:
```python
def greet(name):
    print(f"Hello, {name}!")
greet("Saptanshu")
```

```
Hello, Saptanshu!
```

In [6]:
```python
def add(a,b):
    return a+b

add(1,2)
```

Out[6]: 3

In [12]:
```python
def power(base,exp=2):
    return base**exp

print( power(2) )
print( power(2,5) )
```

```
4
32
```

In [14]:
```python
def factorial(n):
    if n==0 or n==1:
        return 1
    else:
        return n*factorial(n-1)

factorial(5)
```

Out[14]: 120

In [10]:
```
def fibonacci(n):
    if n<=0:
        return []
    if n==1:
        return [0]
    if n==2:
        return [0,1]
    else:
        fib=fibonacci(n-1)
        fib.append(fib[-1]+fib[-2])
        return fib

fibonacci(10)
```

Out[10]: [0, 1, 1, 2, 3, 5, 8, 13, 21, 34]